■ BullzAI — Comprehensive Technical Specification

Institutional-Grade Stock Intelligence, Disambiguation & AI Market Sentiment

Project Version: v0.2.0	**Target Environment:** Google Chrome (MV3)
Author / Engineer: AI Generalist	**Date:** November 2026
Core Stack: TypeScript, Esbuild, Lightweight Charts, Gemini LLM	**Scope:** Indian Financial Markets (NSE/BSE)

1. Executive Summary & Problem Discovery

Financial news readers (investors, traders, researchers) face severe context-switching friction while browsing digital financial news publications (e.g. *The Economic Times*, *Moneycontrol*, *LiveMint*, *Reuters*). When reading about listed companies, users frequently leave articles to manually look up live share prices, search broker terminals, check valuation multiples, and evaluate company news on secondary portals. This research workflow takes 10 to 15 minutes per session and fragments attention.

BullzAI solves this by transforming passive web articles into an interactive, zero-latency financial terminal. It automatically detects company and market index mentions in-situ, overlaying live pricing, institutional 52-week price ranges, interactive multi-timeframe charts, and quantitative AI sentiment verdicts directly within the reader's active workflow.

2. System Architecture & Component Design

BullzAI is architected as a modular Manifest V3 (MV3) Chromium extension. It isolates DOM manipulation, background data orchestration, and client-side UI components to maximize performance and security.

Content Script	src/content/index.ts src/content/scanner.ts	Passively traverses DOM text nodes using optimized Regex lookups. Injects non-intrusive cyber-emerald highlights. Listens for user hover interactions and anchors UI tooltips using pure Shadow DOM.
Disambiguation Engine	src/content/disambiguation.ts	3-tier deterministic filtration system that eliminates ~95% of false-positive detections without incurring costly server or model inference roundtrips.
Background Service Worker	src/background/service-worker.ts	Executes with elevated cross-origin host permissions. Routes market data requests, interacts with unofficial Yahoo Finance endpoints, queries Google News RSS feeds, and coordinates Gemini API queries.

Multi-Tier TTL Cache	src/background/cache.ts	In-memory Maps combined with chrome.storage.session. Quotes cached for 5 min; candlestick data for 15 min; AI news analysis for 45 min to conserve API limits.
UI Components	MiniPopup.ts ExpandedPanel.ts popup.ts	Zero-CSP-conflict Shadow DOM components featuring TradingView canvas charts, 52-week price range sliders, P/E multiples, and live market pulse bars.

3. Detailed Feature Specifications: What & How

3.1 3-Tier Contextual Disambiguation Engine

What it does: Ensures common English words, corporate parent-exchange entities, and ambiguous single-word companies are not mistakenly highlighted when mentioned in non-financial contexts.

How it works: Operates across three deterministic validation gates in disambiguation.ts:

- **Layer 1 (Venue vs. Stock):** Differentiates the exchange platform from the listed stock. Phrases like *'listed on the BSE'* or *'NSE platform'* are rejected, while *'BSE shares rallied 4%'* is highlighted.
- **Layer 2 (Generic Word Gating):** Gated alias table prevents homonyms (e.g. bare adjective *'nifty trick'* is rejected, whereas *'Nifty 50'* or *'Nifty closed higher'* is accepted).
- **Layer 3 (Financial Keyword Proximity):** Tickers in ambiguousBlocklist.json (e.g., *Titan, Route, Force, Star*) inspect a 400-character surrounding paragraph window for financial context tokens (*shares, profit, Q2, dividend, market cap*).

3.2 BullzAI Sentiment Radar (Powered by Google Gemini)

What it does: Automatically converts noisy RSS news feeds into a structured, quantitative investor sentiment gauge (**Bullish / Bearish / Neutral**) with an executive one-sentence verdict.

How it works:

- Fetches candidate headlines from Google News RSS using targeted lookback parameters (48-hour window).
- Formulates a structured JSON schema prompt to Gemini 3.6 Flash requiring sentiment, sentimentScore (40-95%), verdict, and numbered items.
- **Strict Anti-Hallucination Design:** Gemini returns only integer indices referencing the provided headlines. Original URLs, publisher credits, and publication timestamps are preserved directly from the verified RSS feed without model alteration.
- Renders an interactive glowing gauge with color-coded sentiment chips in the right-side overlay panel.

3.3 Institutional Analytics & 52-Week Range Slider

What it does: Provides fundamental valuation ratios and session performance at a single glance.

How it works: Extracts fiftyTwoWeekLow, fiftyTwoWeekHigh, regularMarketDayLow, regularMarketDayHigh, and trailingPE in a single batch quote request. Calculates the relative price position using mathematical interpolation: ((LTP - Low) / (High - Low)) * 100, displaying a calibrated gradient track with a glowing price pin.

3.4 Multi-Timeframe Candlestick Engine (1M, 3M, 6M, 1Y)

What it does: Renders interactive candlestick charts directly within the page DOM without using third-party iframes.

How it works: Employs TradingView's self-hosted lightweight-charts library compiled into the client bundle. Fetches OHLC historical bars via background worker to bypass publisher Content Security Policies (CSP). Users can toggle between **1M, 3M, 6M, and 1Y** intervals, which re-queries and updates chart series via TTL-cached background responses.

3.5 Live Market Pulse & Fuzzy Watchlist Management

What it does: The browser toolbar popup serves as a personal market terminal, syncing saved equities via chrome.storage.sync and tracking major indices in real time.

How it works: Automatically queries live levels for **Nifty 50** (^NSEI) and **Sensex** (^BSESN) upon popup expansion. Incorporates client-side fuzzy searching that filters saved stocks and searches the offline 250+ equity dictionary to discover and add new stocks with a single click.

4. Architectural Tradeoffs & Engineering Decisions

Entity Detection	Deterministic 3-Tier Regex & Dictionary	Full-page LLM Parsing	Eliminates latency (0ms vs 2000ms), removes recurring token API costs, and protects user browsing privacy.
Chart Rendering	Self-hosted Canvas (Lightweight Charts)	Third-party TradingView Iframe Widget	Strict Content Security Policies (CSP) on financial portals (LiveMint, Reuters) block external iframes; Canvas rendering is 100% CSP-immune.
API Routing	Background Worker Proxy	Direct Content-Script Fetching	Host websites enforce strict CORS policies. Manifest V3 service workers with host_permissions bypass CORS restrictions cleanly.
AI Reliability	Index-Referenced Output Schema	Free-form Link & Headline Generation	Guarantees zero hallucinated citations by mapping summaries to strict array indices of verified RSS feed items.

5. Quality Assurance & Verification Rigor

- **Matching Acceptance Tests (npm test):** 32 unit tests verifying longest-alias-first sorting, homonym gating (bare 'Nifty' vs 'Nifty 50'), and venue sense isolation ('BSE-listed' vs 'BSE shares'). Result: **32 Passed, 0 Failed**.
- **Type Safety (npm run typecheck):** Comprehensive TypeScript verification with strict null checks and Chrome MV3 typing. Result: **0 Errors**.
- **End-to-End News Pipeline (npm run verify:news):** Live integration test verifying RSS retrieval, prompt construction, and Gemini AI sentiment calculation.
- **Live Market Index Verification (npm run verify:indices):** Validates that all index ticker symbols return dense candlestick series without data holes.

Conclusion & Impact: *BullzAI exemplifies pragmatic AI engineering: applying machine learning models where they excel (qualitative sentiment synthesis and noise filtering) while relying on high-speed deterministic heuristics for extraction and telemetry. The result is an institutional-grade tool built for speed, reliability, and immediate user adoption.*